The Audit Entropy Theorem: A Second Law for SCX Auditing, the Landauer Cost of Knowledge, and Convergence to Audit Heat Death

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Abstract
SCX $H_A(t) = H(\varepsilon \mid \mathcal{I}_A^{(t)})$ t $\mathcal{I}_A^{(t)}$ ε (i) **??**"" $\mathcal{I}_A^{(t)} \subseteq \mathcal{I}_A^{(t+1)}$ $H_A(t+1) \leq H_A(t)$ $(t+1)$ ε (ii) **??** H_A ΔH $E_{\text{audit}} \geq k_B T \cdot \Delta H \cdot \ln 2$ (iii) **??**$\lim_{t\to\infty} H_A(t) = H_{\min} > 0$ $H_{\min} = H(\varepsilon \mid S)$ 3 [Rigorous](iv) **??**$H_A(1 \oplus 2) \leq \min(H_A(1), H_A(2))$ SCX [Rigorous]

1

$k_B T \ln 2$ [**?**]
SCX

- =
- =
- =
- =

""
AE SCX 3

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(1) **??**[Rigorous]

(2) **??**[Rigorous] / [Open]

(3) **??** $H_{\min} > 0$ 3 [Rigorous]

(4) **??**[Rigorous]

2

2.1

$(\Omega, \mathcal{F}, P)$ $\varepsilon : \Omega \to \{0,1\}$ $\mathcal{F}$-
σ- $\mathcal{G} \subseteq \mathcal{F}$

$$H(\varepsilon \mid \mathcal{G}) = -\mathbb{E}\bigl[\log_2 P(\varepsilon \mid \mathcal{G})\bigr],$$

$\log_2$

2.2

Definition 2.1 (). *t $\mathcal{I}_A^{(t)}$ $\mathcal{F}$ σ- t Cercis Hessian*

$$H_A(t) = H\bigl(\varepsilon \mid \mathcal{I}_A^{(t)}\bigr) = -\mathbb{E}\Bigl[\log_2 P\bigl(\varepsilon \mid \mathcal{I}_A^{(t)}\bigr)\Bigr], \tag{1}$$

t $\varepsilon(x) \in \{0,1\}$

$H_A(t)$ $H_A = 0$ 3

Definition 2.2 ().

$$\mathcal{I}_A^{(t)} \subseteq \mathcal{I}_A^{(t+1)}, \qquad \forall t \geq 0, \tag{2}$$

σ- $\mathcal{I}_A^{(t)}$ $\mathcal{F}$ *σ-filtration*

2.3

Assumption 2.3 (). $(\Omega, \mathcal{F}, P)$ $\varepsilon \in L^1(\Omega)$ $\mathbb{E}[|\varepsilon|] < \infty$

Assumption 2.4 (). $\{\mathcal{I}_A^{(t)}\}_{t \geq 0}$ $\mathcal{F}$ *filtration* $\mathcal{I}_A^{(t)} \subseteq \mathcal{I}_A^{(t+1)} \subseteq \mathcal{F}$ t

Assumption 2.5 (SCX). *tCercis S ∇S Hessian H_S $\mathcal{I}_A^{(t)}$-$S \in \mathcal{I}_A^{(\infty)}$ $\mathcal{I}_A^{(\infty)} := \sigma\bigl(\bigcup_{t=0}^{\infty} \mathcal{I}_A^{(t)}\bigr)$*

Assumption 2.6 (3). *SCX σ- $\mathfrak{G}_{\mathrm{SCX}} = \sigma(\{S, \nabla S, H_S\})$ 3*

$$H\bigl(\varepsilon \mid \mathfrak{G}_{\mathrm{SCX}}\bigr) = H(\varepsilon \mid S) > 0.$$

$\mathfrak{G}_{\mathrm{SCX}}$ ε σ- σ- $\mathcal{H} \subseteq \mathcal{F}$ $S \in \mathcal{H}$ $\mathcal{H} \subseteq \mathfrak{G}_{\mathrm{SCX}}$ $H(\varepsilon \mid \mathcal{H}) \geq H(\varepsilon \mid \mathfrak{G}_{\mathrm{SCX}})$

2.4

$\mathcal{I}_A^{(t)}$

(i) **Cercis** $S(x) = Q(x) + \eta N(x)$

(ii) $J_k(x) \in \{\text{noise}, \text{hard}\}$

(iii)

(iv)

ε

3

3.1

Theorem 3.1 (). **??** **??** (**??**) $\mathcal{A}$

$$H_A(t+1) \;\le\; H_A(t), \tag{3}$$

$$I\big(\varepsilon; \mathcal{I}_A^{(t+1)} \mid \mathcal{I}_A^{(t)}\big) = 0,$$

$(t+1)$ ε $\varepsilon \perp\!\!\!\perp \mathcal{I}_A^{(t+1)} \mid \mathcal{I}_A^{(t)}$

AE.1 . $\mathcal{I}_t := \mathcal{I}_A^{(t)}$ σ-

$$H(\varepsilon \mid \mathcal{I}_{t+1}) = H(\varepsilon \mid \mathcal{I}_t, \mathcal{I}_{t+1}),$$

$\mathcal{I}_{t+1}$ $\mathcal{I}_t$

$\mathcal{I}_t \subseteq \mathcal{I}_{t+1}$ **??** σ- $\sigma(\mathcal{I}_{t+1}) \supseteq \sigma(\mathcal{I}_t)$ σ- $\mathcal{G}_1 \subseteq \mathcal{G}_2$ σ-

$$H(\varepsilon \mid \mathcal{G}_1) \ge H(\varepsilon \mid \mathcal{G}_2).$$

$\mathcal{G}_1 = \sigma(\mathcal{I}_t)$ $\mathcal{G}_2 = \sigma(\mathcal{I}_{t+1})$

$$H(\varepsilon \mid \mathcal{I}_t) \ge H(\varepsilon \mid \mathcal{I}_{t+1}),$$

(**??**)

$$\Delta I_t := I(\varepsilon; \mathcal{I}_{t+1} \mid \mathcal{I}_t) = H(\varepsilon \mid \mathcal{I}_t) - H(\varepsilon \mid \mathcal{I}_t, \mathcal{I}_{t+1}).$$

$$H(\varepsilon \mid \mathcal{I}_t) = H(\varepsilon \mid \mathcal{I}_{t+1}) + \Delta I_t.$$

$H(\varepsilon \mid \mathcal{I}_{t+1}) = H(\varepsilon \mid \mathcal{I}_t)$ $\Delta I_t = 0$

$\Delta I_t = 0$ $\varepsilon \perp\!\!\!\perp \mathcal{I}_{t+1} \mid \mathcal{I}_t$ $\mathcal{I}_{t+1}$ ε $\mathcal{I}_t$

$$P(\varepsilon \in A \mid \mathcal{I}_t, \mathcal{I}_{t+1}) = P(\varepsilon \in A \mid \mathcal{I}_t), \quad \forall A \in \mathcal{B}(\{0,1\}).$$

ΔI_t

$$\begin{aligned} \Delta I_t &= I(\varepsilon; \mathcal{I}_{t+1} \mid \mathcal{I}_t) \\ &= H(\mathcal{I}_{t+1} \mid \mathcal{I}_t) - H(\mathcal{I}_{t+1} \mid \varepsilon, \mathcal{I}_t) \\ &= \mathbb{E}\Big[\log_2 \frac{P(\mathcal{I}_{t+1}, \varepsilon \mid \mathcal{I}_t)}{P(\mathcal{I}_{t+1} \mid \mathcal{I}_t) P(\varepsilon \mid \mathcal{I}_t)}\Big]. \end{aligned}$$

$P(\mathcal{I}_{t+1}, \varepsilon \mid \mathcal{I}_t) = P(\mathcal{I}_{t+1} \mid \mathcal{I}_t) P(\varepsilon \mid \mathcal{I}_t)$ $\Delta I_t = 0$□ □

?? SCX (**??**) $H_A(t)$ $H_A(t)$ $H_A(t+1) = H_A(t)$

Remark 3.2 (). **??** *DPI [**?**][Rigorous]*

Remark 3.3 (). *[Honest Critique] AE.1*

1. $\mathcal{I}_A^{(t+1)} \not\supseteq \mathcal{I}_A^{(t)}$
2. $\mathcal{I}_A^{(t)} \subseteq \mathcal{I}_A^{(t+1)}$ σ-

3.2

Theorem 3.4 (). $\Delta H = H_A(t) - H_A(t+1) > 0$

$$E_{\text{audit}} \;\geq\; k_B T \cdot \Delta H \cdot \ln 2, \tag{4}$$

k_B T

AE.2 . $H_A(t) = H(\varepsilon \mid \mathcal{I}_A^{(t)})$ **??** t $2^{H_A(t)}$

ρ_t t $W(t) = e^{H_A(t)\ln 2} = 2^{H_A(t)}$

$$S_{\text{phys}}(t) = k_B \ln W(t) = k_B H_A(t) \ln 2.$$

$\Delta H = H_A(t) - H_A(t+1)$

$$\Delta S_{\text{phys}} = k_B (\ln 2)\big(H_A(t) - H_A(t+1)\big) = k_B \Delta H \ln 2.$$

[**?**] $k_B T \ln 2$ ΔH ΔS_{phys}

ΔQ

$$\Delta S = \frac{\Delta Q}{T} \geq \Delta S_{\text{phys}}.$$

$$E_{\text{audit}} \geq \Delta Q \geq T \cdot \Delta S_{\text{phys}} = k_B T \cdot \Delta H \cdot \ln 2.$$

nats **bits** nats bits $H_A^{\text{nats}} = H_A^{\text{bits}} \cdot \ln 2$

$$\Delta H^{\text{bits}} = \frac{\Delta H^{\text{nats}}}{\ln 2}.$$

$$E_{\text{audit}} \geq k_B T \cdot \frac{\Delta H^{\text{nats}}}{\ln 2} \cdot \ln 2 = k_B T \cdot \Delta H^{\text{nats}}.$$

$E_{\text{audit}} \geq k_B T \cdot (\text{ nats }) = k_B T \cdot (\text{ bits }) \cdot \ln 2\square$ $\square$

?? ΔH $k_B T \cdot \Delta H \cdot \ln 2$ 10^9 $\Delta H = 0.1$ 0.5 0.4 2.8×10^{-13} 2.8×10^{-4} ?? SCX

Open Problem 3.5 (). *$E_{\text{audit}} = k_B T \cdot \Delta H \cdot \ln 2$ Situs Situs [Open]*

Remark 3.6 (). *?? [?][Conditionally Rigorous]*

Remark 3.7 (). *[Honest Critique] AE.2*

1. *ALU [?]*
   (a) (b) ΔH
2. *Bennett pebble game [?] ΔH*
3. *H_A H_A*

3.3

Theorem 3.8 (). *?????? ??*

$$\lim_{t \to \infty} H_A(t) \;=\; H_{\min} \;>\; 0, \tag{5}$$

$$H_{\min} = H(\varepsilon \mid S) = H(\varepsilon) - I(\varepsilon; S) \tag{6}$$

ε 3

AE.3 .
$\{H_A(t)\}_{t=0}^{\infty}$

i AE.1$H_A(t+1) \leq H_A(t)$ $t \geq 0$

ii $H_A(t) = H(\varepsilon \mid \mathcal{I}_A^{(t)}) \geq 0$

$$\lim_{t\to\infty} H_A(t) =: H_{\min} \qquad H_{\min} \geq 0.$$

σ-

$$\mathcal{I}_A^{(\infty)} := \sigma\Big(\bigcup_{t=0}^{\infty} \mathcal{I}_A^{(t)}\Big).$$

$t \to \infty$ $P(\varepsilon \mid \mathcal{I}_A^{(t)}) \to P(\varepsilon \mid \mathcal{I}_A^{(\infty)})$ a.s. $H(\varepsilon \mid \mathcal{I}_A^{(t)}) \to H(\varepsilon \mid \mathcal{I}_A^{(\infty)})$

i **??**Cercis S $\mathcal{I}_A^{(\infty)}$- $\sigma(S) \subseteq \mathcal{I}_A^{(\infty)}$

ii DPI$\sigma(S) \subseteq \mathcal{I}_A^{(\infty)}$

$$H_{\min} = H\big(\varepsilon \mid \mathcal{I}_A^{(\infty)}\big) \leq H(\varepsilon \mid S).$$

iii **??**SCX σ- $\mathfrak{G}_{\mathrm{SCX}} = \sigma(\{S, \nabla S, H_S\})$ ε $\mathcal{I}_A^{(\infty)}$ $\mathcal{I}_A^{(\infty)} \subseteq \mathfrak{G}_{\mathrm{SCX}} \vee \mathcal{N}$ $\mathcal{I}_A^{(\infty)} \subseteq \mathfrak{G}_{\mathrm{SCX}}$ DPI

$$H_{\min} = H\big(\varepsilon \mid \mathcal{I}_A^{(\infty)}\big) \geq H\big(\varepsilon \mid \mathfrak{G}_{\mathrm{SCX}}\big).$$

iv T3 **??**

$$H\big(\varepsilon \mid \mathfrak{G}_{\mathrm{SCX}}\big) = H(\varepsilon \mid S) > 0.$$

iiiii

$$H(\varepsilon \mid S) \leq H_{\min} \leq H(\varepsilon \mid S) \quad \Longrightarrow \quad H_{\min} = H(\varepsilon \mid S).$$

v 3$H(\varepsilon \mid S) > 0$ $H_{\min} > 0$

$H_{\min} = H(\varepsilon \mid S)$

$$\lim_{t\to\infty} H_A(t) = H_{\min} = H(\varepsilon \mid S) > 0.$$

σ-$\mathcal{I}_A^{(\infty)}$ $\mathfrak{G}_{\mathrm{SCX}}$ SCX T3

$$\mathcal{I}_A^{(\infty)} = \mathfrak{G}_{\mathrm{SCX}} \quad (),$$

□ □

?? SCX $H_{\min} = H(\varepsilon \mid S) > 0$ SLA $1 - H_{\min}/\log 2$ 100% $H_{\min}$ Cercis SCX $H_{\min}$ RA-Theorem η^* **??** $H_{\min}$

Remark 3.9. *“” SCX 3* $H_{\min} > 0$*[Rigorous]*

Remark 3.10 (). *[Honest Critique]* *AE.3*

1. ***T3*** *σ-* *AE.3* $\mathcal{I}_A^{(\infty)} \subseteq \mathfrak{G}_{\mathrm{SCX}}$ *SCX* ε $\mathfrak{G}_{\mathrm{SCX}}$ J_k *SCX* $\sigma(J_k) \nsubseteq \mathfrak{G}_{\mathrm{SCX}}$ $H_{\min}$

   $H_{\min}$ SCX $H_{\min}$

2. $t \to \infty$ $tH_A(t) - H_{\min}$

3.

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   -
   -

   “”“”

3.4

Theorem 3.11 (). A_1 A_2 $\mathcal{I}_1$ $\mathcal{I}_2$ $H_A(1) = H(\varepsilon \mid \mathcal{I}_1)$ $H_A(2) = H(\varepsilon \mid \mathcal{I}_2)$ $\mathcal{I}_1 \cup \mathcal{I}_2$

$$H_A(1 \oplus 2) = H(\varepsilon \mid \mathcal{I}_1, \mathcal{I}_2) = H(\varepsilon \mid \mathcal{I}_1) + H(\varepsilon \mid \mathcal{I}_2) - H(\varepsilon) - I(\mathcal{I}_1; \mathcal{I}_2 \mid \varepsilon) + I(\mathcal{I}_1; \mathcal{I}_2). \tag{7}$$

$$H_A(1 \oplus 2) \;\leq\; \min\bigl(H_A(1),\, H_A(2)\bigr), \tag{8}$$

$\varepsilon \perp\!\!\!\perp \mathcal{I}_2 \mid \mathcal{I}_1$ $\varepsilon \perp\!\!\!\perp \mathcal{I}_1 \mid \mathcal{I}_2$

$$H_A(1 \oplus 2) \;\geq\; H_A(1) + H_A(2) - H(\varepsilon) - I(\mathcal{I}_1; \mathcal{I}_2 \mid \varepsilon), \tag{9}$$

$I(\mathcal{I}_1; \mathcal{I}_2) = 0$ $\mathcal{I}_1$ $\mathcal{I}_2$
$\mathcal{I}_1 \perp\!\!\!\perp \mathcal{I}_2 \mid \varepsilon$ ε $I(\mathcal{I}_1; \mathcal{I}_2 \mid \varepsilon) = 0$

$$H_A(1 \oplus 2) = H_A(1) + H_A(2) - H(\varepsilon) + I(\mathcal{I}_1; \mathcal{I}_2). \tag{10}$$

AE.4 .

$$\begin{aligned} H_A(1 \oplus 2) &= H(\varepsilon \mid \mathcal{I}_1, \mathcal{I}_2) \\ &= H(\varepsilon \mid \mathcal{I}_1) - I(\varepsilon; \mathcal{I}_2 \mid \mathcal{I}_1) \\ &= H(\varepsilon \mid \mathcal{I}_2) - I(\varepsilon; \mathcal{I}_1 \mid \mathcal{I}_2) \quad . \end{aligned}$$

$I(\varepsilon;\mathcal{I}_2 \mid \mathcal{I}_1)$

$$\begin{aligned} I(\varepsilon;\mathcal{I}_2 \mid \mathcal{I}_1) &= H(\mathcal{I}_2 \mid \mathcal{I}_1) - H(\mathcal{I}_2 \mid \varepsilon,\mathcal{I}_1) \\ &= \big[H(\mathcal{I}_2) - I(\mathcal{I}_1;\mathcal{I}_2)\big] - \big[H(\mathcal{I}_2 \mid \varepsilon) - I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon)\big] \\ &= \big[H(\mathcal{I}_2) - H(\mathcal{I}_2 \mid \varepsilon)\big] - I(\mathcal{I}_1;\mathcal{I}_2) + I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon) \\ &= I(\varepsilon;\mathcal{I}_2) - I(\mathcal{I}_1;\mathcal{I}_2) + I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon). \end{aligned}$$

$$\begin{aligned} H_A(1 \oplus 2) &= H(\varepsilon \mid \mathcal{I}_1) - \big[I(\varepsilon;\mathcal{I}_2) - I(\mathcal{I}_1;\mathcal{I}_2) + I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon)\big] \\ &= H(\varepsilon \mid \mathcal{I}_1) - I(\varepsilon;\mathcal{I}_2) + I(\mathcal{I}_1;\mathcal{I}_2) - I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon). \end{aligned}$$

$$H(\varepsilon) - I(\varepsilon;\mathcal{I}_2) = H(\varepsilon \mid \mathcal{I}_2) = H_A(2) \ I(\varepsilon;\mathcal{I}_2) = H(\varepsilon) - H_A(2)$$

$$\begin{aligned} H_A(1 \oplus 2) &= H_A(1) - \big[H(\varepsilon) - H_A(2)\big] + I(\mathcal{I}_1;\mathcal{I}_2) - I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon) \\ &= H_A(1) + H_A(2) - H(\varepsilon) + I(\mathcal{I}_1;\mathcal{I}_2) - I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon). \end{aligned}$$

(**??**)

$$H_A(1 \oplus 2) = H(\varepsilon \mid \mathcal{I}_1) - I(\varepsilon;\mathcal{I}_2 \mid \mathcal{I}_1) \le H(\varepsilon \mid \mathcal{I}_1) = H_A(1),$$

$H_A(1 \oplus 2) \le H_A(2)$

$$H_A(1 \oplus 2) \le \min(H_A(1), H_A(2)).$$

$I(\varepsilon;\mathcal{I}_2 \mid \mathcal{I}_1) = 0$ $H_A(1)$ $\varepsilon \perp\!\!\!\perp \mathcal{I}_2 \mid \mathcal{I}_1 \varepsilon \perp\!\!\!\perp \mathcal{I}_1 \mid \mathcal{I}_2$ $H_A(1 \oplus 2) = H_A(2)$
(**??**) $I(\mathcal{I}_1;\mathcal{I}_2) \ge 0$

$$H_A(1 \oplus 2) = H_A(1) + H_A(2) - H(\varepsilon) - I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon) + I(\mathcal{I}_1;\mathcal{I}_2) \ge H_A(1) + H_A(2) - H(\varepsilon) - I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon).$$

$I(\mathcal{I}_1;\mathcal{I}_2) = 0$
$\mathcal{I}_1 \perp\!\!\!\perp \mathcal{I}_2 \mid \varepsilon$ $I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon) = 0$ (**??**)

$$H_A(1 \oplus 2) = H_A(1) + H_A(2) - H(\varepsilon) + I(\mathcal{I}_1;\mathcal{I}_2).$$

$\mathcal{I}_1$ $\mathcal{I}_2$ ε □ □

?? SCX (**??**) $I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon)$ Cercis FA-Theorem$\mathcal{I}_1 \approx \mathcal{I}_2$ $\mathcal{I}_1 \perp \mathcal{I}_2 \mid \varepsilon$ SCX $I(\mathcal{I}_k;\mathcal{I}_{\text{others}} \mid \varepsilon)$

Remark 3.12 (). **??** (**??**) *[Rigorous]*

Remark 3.13 (). *[Honest Critique] AE.4*

1. $I(\mathcal{I}_1;\mathcal{I}_2) = 0$ $\mathcal{I}_1$ $\mathcal{I}_2$ *SCX* $\mathcal{I}_1$ $\mathcal{I}_2$ ε ε $I(\mathcal{I}_1;\mathcal{I}_2) > 0$
2. $\mathcal{I}_1 \perp \mathcal{I}_2 \mid \varepsilon$ ε ε $I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon) > 0$
3. $I(\mathcal{I}_1;\mathcal{I}_2 \mid \varepsilon)$ $I(\mathcal{I}_1;\mathcal{I}_2)$

4 SCX

SCX

	AE-Theorem	
$dS \geq 0$	$dH_A \leq 0$	[Rigorous]
		[Rigorous] / [Open]
$S \to S_{\max}$	$H_A \to H_{\min} > 0$	[Rigorous]
		[Rigorous]

4.1

H_A H_A $\mathcal{I}_A^{(t)}$

4.2 RA-Theorem

$H_{\min}$ η^*RA-Theorem [?] $H_{\min} = H(\varepsilon \mid S) = \eta^*$ 3

5

AE-Theorem SCX

(1) H_A [Rigorous]

(2) $E_{\text{audit}} \geq k_B T \cdot \Delta H \cdot \ln 2$[Rigorous] / [Open]

(3) $\lim H_A = H_{\min} > 0$[Rigorous]

(4) $H_A(1 \oplus 2) = H_A(1) + H_A(2) - H(\varepsilon) + I(\mathcal{I}_1; \mathcal{I}_2) - I(\mathcal{I}_1; \mathcal{I}_2 \mid \varepsilon) \leq \min(H_A(1), H_A(2))$[Rigorous]

?? SCX SCX

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